

Differentiation test · Algebra

Grade 11

40 minutes 100 marks 15 questions + bonus No calculator

Lessons 3–4, 7–12 and 17–18

- Answer on separate paper, and show your working — method carries marks.
- Simplify every derivative. Leave no negative or fractional index in a final answer.

NAME

CLASS

DATE

MARK / 100

A

Section A · Five marks each

EASY 40 marks

Differentiate and simplify. One line of working is enough.

- 1

$y = 4x^3 - 5x + 7$ [5]
- 2

$y = \frac{3}{x^2}$ [5]
- 3

$y = 5\sqrt{x}$ [5]
- 4

$y = \frac{x^3 - 2x}{x}$ [5]
- 5

$y = (3x + 1)^5$ [5]
- 6

$y = (x^2 - 4)^3$ [5]
- 7

$y = \frac{1}{2x - 5}$ [5]
- 8

$y = \sqrt{4x + 1}$ [5]

B

Section B · Eight marks each

MEDIUM 40 marks

Method marks are available even when the final answer is wrong.

- 9

Differentiate $y = \frac{2x - 1}{x^2 + 3}$ and find the gradient of the curve at $x = 1$. [8]
- 10

$y = x^3 - 3x^2 + 4$. Find the values of x at which the gradient of the curve is 9. [8]
- 11

$y = \sqrt{x^2 + 9}$. Find $\frac{dy}{dx}$ and its value at $x = 4$. [8]

12 Differentiate $y = \frac{5}{(1-2x)^3}$. [8]

13 Evaluate:

(a) $\lim_{x \rightarrow 3} \frac{x^2 - 9}{x - 3}$ [4]

(b) $\lim_{x \rightarrow 0} \frac{\sqrt{x+4} - 2}{x}$ [4]

C Section C · Ten marks each

Full method required.

HARD

20 marks

14 $y = x^2\sqrt{2x-1}$.

(a) Show that $\frac{dy}{dx} = \frac{x(5x-2)}{\sqrt{2x-1}}$. [6]

(b) Hence find the gradient of the curve at $x = 5$. [4]

15 The curve $y = ax^2 + bx$ passes through (2, 2) and has gradient 5 at $x = 2$.

(a) Find a and b . [7]

(b) Hence find the equation of the tangent to the curve at (2, 2). [3]

Bonus



Attempt this only when the rest is finished. It can make up marks lost elsewhere, but your total is still recorded out of 100.

BONUS

+10 marks

16 $y = \frac{x+1}{\sqrt{x}}$ for $x > 0$.

(a) Show that $\frac{dy}{dx} = \frac{x-1}{2x\sqrt{x}}$. [7]

(b) Hence find the gradient of the curve at $x = 4$. [3]